Mapping O*NET Tasks to ISCO Occupations using Text Similarity

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Abstract

O*NET task data are widely used to measure technology exposure and skill demand, but O*NET is based on the U.S. Standard Occupational Classification (SOC) while international labour statistics use ISCO. Existing SOC–ISCO concordances operate at the occupation level, ignoring within-occupation task variation. This paper constructs a task-level crosswalk from O*NET to 4-digit ISCO-08 occupations using text similarity and ESCO, assigning 18,796 tasks to 435 of 436 ISCO-08 groups. Validation against institutional crosswalks shows 68.0% exact agreement and 88.4% major-group agreement. The crosswalk enables task-based measures of automation, skills, and AI exposure to be applied to internationally comparable labour statistics.

Keywords: Occupational classification; O*NET; ISCO-08; Task-level crosswalk; Text similarity; Labour market

JEL codes: J24; J44; C88; O33; C80

1 Introduction

Many studies in labour economics use O*NET task data to measure technology exposure, skill demand, and occupational change (Autor et al., 2003; Acemoglu and Autor, 2011; Frey and Osborne, 2017; Acemoglu and Restrepo, 2020; Atalay et al., 2020). O*NET provides detailed descriptions of job tasks and has become a standard source for analysing how technological change affects the nature of work. More recent datasets measuring exposure to artificial intelligence and digital technologies are also published at the O*NET task level (Eloundou et al., 2024; Shao et al., 2025; Handa et al., 2025; Appel, McCrory, et al., 2025; Appel, Massenkoff, et al., 2026; Massenkoff et al., 2026), based on both the SOC 2010 and SOC 2018 versions of O*NET.

O*NET is organised under the U.S. Standard Occupational Classification (SOC), while most international employment statistics are indexed using the International Standard Classification of Occupations (ISCO) (International Labour Office, 2012). Existing SOC–ISCO concordances operate at the occupation level (ESCO Secretariat, 2022; Bureau of Labor Statistics, 2015; Matysiak et al., 2024). These mappings allow researchers to translate occupation-level measures across classification systems, but they ignore differences in tasks within occupations. Task-level information is central to much of the recent literature on technological change, where

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exposure to automation or artificial intelligence is measured using detailed descriptions of work activities rather than occupational titles alone. Aggregating tasks to occupations before applying a crosswalk removes this within-occupation variation and reduces the precision of empirical analysis.

This paper builds a direct task-level crosswalk from O*NET to 4-digit ISCO-08 occupations using text similarity. The method compares the text of O*NET task statements with occupational descriptions from ISCO-08 and ESCO, allowing each task to be matched to the most similar ISCO-08 occupation group. Two crosswalk versions are provided, corresponding to O*NET releases based on SOC 2018 and SOC 2010 classifications, allowing compatibility with both recent and earlier task-based datasets.

The resulting crosswalk assigns 18,796 O*NET tasks to 435 of 436 ISCO-08 unit groups. Validation against existing SOC–ISCO concordances shows 68.0% exact agreement and 88.4% agreement at the major-group level. The mapping is also highly stable across O*NET releases. Among the 16,049 task IDs shared between O*NET 29.2 (SOC 2018) and O*NET 25.0 (SOC 2010), top-link ISCO assignments agree in 98.1% of cases. The crosswalk allows existing task-based measures of automation, skills, and AI exposure to be applied directly to internationally comparable labour statistics.

Section 2 describes the data sources. Section 3 outlines the method. Section 4 presents validation results. Section 5 discusses parameter choices. Section 6 concludes. All appendices are provided in the online supplementary material.

2 Data

2.1 Tasks and Occupations data

The primary task data source is O*NET 29.2, the current release under the SOC 2018 taxonomy (Office of Management and Budget, 2018), which contains 18,796 task statements keyed by official task ID. Each statement describes a discrete work activity performed within a specific SOC occupation, and the parent occupation title is used as additional context in the matching procedure. A secondary release is built from O*NET 25.0, the final release under SOC 2010, which contains 19,735 task statements and enables application to datasets constructed using the earlier taxonomy, including early versions of the Anthropic Economic Index (Appel, McCrory, et al., 2025; Appel, Massenkoff, et al., 2026). O*NET is a standard source of occupational task information in the labour economics literature; see Peterson et al. (2001) for a detailed description.

The target classification is ISCO-08 at the four-digit unit-group level (International Labour Office, 2012). ISCO-08 is the international standard used in most cross-country labour statistics.

I also use data from ESCO (European Commission, 2019). ESCO includes approximately 3,000 occupation concepts linked to ISCO-08 unit groups and adds skill and competence information that complements the official ISCO-08 task descriptions.

2.2 Institutional crosswalks for validation

I evaluate the crosswalk using four existing SOC–ISCO-08 concordances. These crosswalks are used only for validation and do not affect the construction of the task-level mapping.

For SOC 2018, I use two concordances linking ESCO and O*NET occupations: the ESCO–SOC 2018 crosswalk (3,834 occupation pairs) (O*NET Center, 2022) and the reverse mapping, the SOC 2018–ESCO crosswalk (2,756 pairs) (ESCO Secretariat, 2022).

For SOC 2010, I use the official U.S. Bureau of Labor Statistics SOC 2010–ISCO-08 crosswalk (1,112 pairs) (Bureau of Labor Statistics, 2015) and an ESCO–O*NET crosswalk with semantic similarity scores (1,680 pairs) (Matysiak et al., 2024).

Agreement across these sources provides an external benchmark for assessing the consistency of the task-level assignments. Section 4 reports agreement rates under alternative definitions of acceptable matches.

3 Method

3.1 Overview

The pipeline assigns each O*NET task statement to exactly one ISCO-08 unit group using text similarity. Figure 1 summarises the data flow. Each task is represented using its text description, associated DWA labels, and its parent SOC occupation title. Each ISCO-08 occupation is represented using official ISCO-08 task descriptions enriched with ESCO occupation text and associated skill information.

All texts are encoded with the `all-mpnet-base-v2` sentence embeddings (Reimers and Gurevych, 2019). Cosine similarity between task and occupation representations determines matches. The assignment procedure selects the most similar ISCO-08 group for each task, with a coverage step ensuring representation of all occupied unit groups.

The approach preserves task-level variation within occupations and allows task-based measures derived from O*NET to be translated to internationally comparable labour statistics.

3.2 Task and occupation representations

Each O*NET task is represented using the task statement text, associated Detailed Work Activity (DWA) labels, and the title of the SOC occupation to which the task belongs. Sentence embeddings are computed for all three components in a two-step blend. DWA labels are first combined with the task text,

$$e_{\text{core}} = (1 - w_{\text{dwa}})e_{\text{task}} + w_{\text{dwa}}e_{\text{DWA}}, \tag{1a}$$

and the result is then blended with the SOC occupation title,

$$q = (1 - w_{\text{soc}})e_{\text{core}} + w_{\text{soc}}e_{\text{SOC}}, \tag{1b}$$

where e_{task} denotes the embedding of the task text, e_{DWA} the embedding of associated DWA labels, and e_{SOC} the embedding of the SOC occupation title. With the selected parameters ($w_{\text{dwa}} = 0.266$, $w_{\text{soc}} = 0.375$), the effective weights are approximately 45.9% task text, 16.6% DWA labels, and 37.5% SOC title. Including DWA labels enriches task representations with activity-level specificity; including occupation titles helps distinguish tasks that use similar language but occur in different occupational contexts.

Each ISCO-08 occupation is represented using both official ISCO-08 task descriptions and ESCO occupation information. The ISCO component combines task descriptions with summary occupation text,

$$e_{\text{ISCO}} = w_{\text{isco,task}} e_{\text{tasks}} + (1 - w_{\text{isco,task}}) e_{\text{info}}. \tag{2}$$

The ESCO component combines occupation descriptions with associated skill text,

$$e_{\text{ESCO}} = w_{\text{occ}} e_{\text{occ}} + (1 - w_{\text{occ}}) e_{\text{skills}}. \tag{3}$$

Where multiple ESCO occupations map to the same ISCO-08 unit group, their embeddings are averaged.

The final occupation representation is a weighted combination of ISCO and ESCO information,

$$e_{\text{target}} = w_{\text{isco}} e_{\text{ISCO}} + (1 - w_{\text{isco}}) e_{\text{ESCO}}. \tag{4}$$

In implementation, component vectors are L2-normalised before blending, and the blended query and final target vectors are L2-normalised before retrieval; weights are nominal blend weights.

Combining ISCO-08 task descriptions with ESCO occupation and skill text improves coverage and provides richer context on occupational tasks and required skills.

3.3 Assignment rule

For each task representation, cosine similarity is computed against all ISCO-08 occupation representations. The top-ranked candidate for each task is always retained as its assignment. The similarity threshold (`min_sim` = 0.45) and margin (`margin_best` = 0.03) govern whether any additional near-best candidates are kept; with `max_links_per_task` = 1, no additional links are added in the main specification, so each task receives exactly one ISCO-08 assignment.

A coverage backfill step assigns any tasks not yet matched by the primary filter to ISCO-08 unit groups not otherwise covered, ensuring every occupation group is represented in the crosswalk. This supports aggregation of task-level measures to occupation-level statistics.

Parameter values are selected using an unsupervised search procedure described in Section 5.

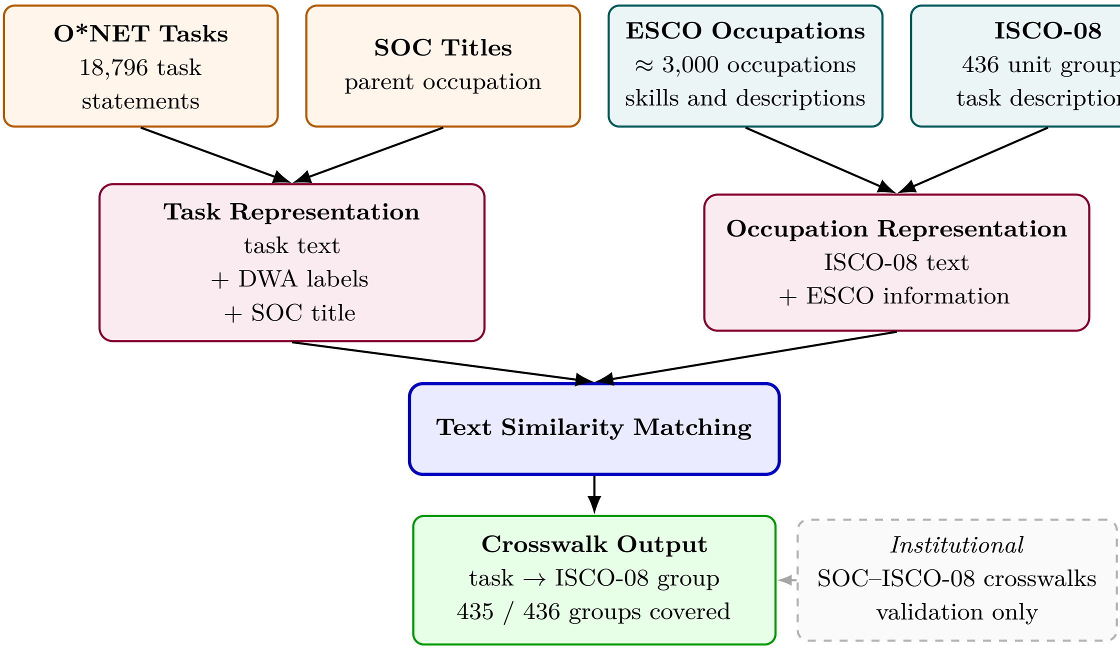

Figure 1: Task-level mapping from O*NET to ISCO-08.

Notes: Task representations combine O*NET task text, DWA labels, and SOC occupation context. Occupation representations combine official ISCO-08 task descriptions with ESCO occupation text and skill information. Similarity matching assigns each task to exactly one ISCO-08 unit group. Institutional SOC–ISCO-08 crosswalks are used only for validation.

4 Validation

Because no gold standard exists for mapping tasks directly to ISCO-08 occupations, validation is performed indirectly using institutional occupation-level SOC–ISCO crosswalks and additional qualitative assessment. The task-level mapping is first aggregated to the occupation level, allowing comparison with existing concordances constructed by statistical agencies and international organisations. These crosswalks provide an external benchmark for evaluating whether semantic similarity between task and occupation text produces assignments consistent with established classification correspondences.

Agreement with institutional SOC–ISCO crosswalks is high. For the O*NET 29.2 (SOC 2018) version, 68.0% of SOC occupations are mapped to the same ISCO-08 unit group as in at least one institutional crosswalk, while 88.4% agree at the ISCO major-group level.

Cross-release stability is evaluated on the 16,049 task IDs appearing in both releases. These IDs mostly preserve task text, but not always SOC occupation context because O*NET 29.2 uses SOC 2018 while O*NET 25.0 uses SOC 2010. Top-link ISCO assignments agree for 98.1% of shared task IDs.

Agreement rates vary only slightly across alternative definitions of the reference set based on individual crosswalks or their union. Detailed comparison tables are provided in Online Appendix C. These results indicate that the task-level mapping reproduces most occupation-level correspondences identified in institutional concordances, while allowing differences when task descriptions suggest alternative matches.

Results are broadly stable across parameter choices; cross-release top-link stability is high, although institutional validation rates differ between O*NET releases. Sensitivity analysis varying embedding models, component weights, and similarity thresholds produces similar agreement rates, indicating that the mapping is not driven by specific parameter settings. Online Appendix A reports agreement statistics across alternative specifications. The version constructed using O*NET 25.0 (SOC 2010) uses the same specification and shows high top-link stability across shared task IDs, although institutional validation rates are lower for the SOC 2010 version; the crosswalk supports compatibility with datasets based on earlier O*NET releases, including early versions of the Anthropic Economic Index.

Qualitative inspection of sampled tasks confirms that matched occupations are generally semantically consistent with the underlying task descriptions. Mismatches occur primarily in cases where task descriptions are very broad or where occupational boundaries differ between U.S. and international classification systems. Examples of task assignments are provided in Online Appendix E. Author annotation of a purposive sample of 108 O*NET29.2 tasks (36 occupations, stratified by ISCO major group) confirms this pattern: the selected configuration achieves 36.1% exact four-digit agreement, rising to 72.2% at the major-group level (Online Appendix D). Overall, the results indicate that text similarity provides a consistent and transparent basis for constructing task-level correspondences between O*NET and ISCO-08.

5 Parameter Selection

The main specification uses weight $w_{\text{soc}} = 0.375$ for the SOC occupation title in the task representation, and weight $w_{\text{isco}} = 0.375$ for the ISCO component of the target representation; within that component, $w_{\text{isco,task}} = 0.734$ weights the ISCO-08 task descriptions. The similarity threshold is set to 0.45 with margin parameter 0.03. Each task receives exactly one ISCO-08 assignment.

Parameter values are chosen using unsupervised criteria reflecting the objectives of the mapping. Candidate specifications are evaluated based on coverage of ISCO-08 unit groups, mean similarity between matched texts, and the distribution of tasks across occupation groups. Configurations that concentrate a large share of tasks in a small number of occupations are penalised, as this indicates weak differentiation in task content.

Three qualitative patterns emerge. First, including Detailed Work Activity labels in the task representation improves performance across evaluation dimensions; DWA labels add activity-level specificity that enriches the semantic content of task embeddings beyond what the task statement alone captures. Second, a moderate weight on the SOC occupation title ($w_{\text{soc}} = 0.375$) produces more stable assignments than either heavier or lighter reliance on occupational context. Third, combining ISCO-08 task descriptions with ESCO occupation and skill text improves retrieval relative to using either source separately, indicating that the two sources provide complementary information.

The selected specification assigns tasks to 435 of the 436 ISCO-08 unit groups, with ISCO 7516 structurally unmatched due to the absence of a functional equivalent in O*NET. Sensitivity analysis reported in Online Appendix A shows that results are stable across reasonable variation in parameter values.

6 Conclusion

This paper presents a semantic crosswalk that maps 18,796 task statements from O*NET 29.2 to ISCO-08 four-digit unit groups, covering 435 of the 436 groups in the classification. The approach combines task text with occupational context and enriched ISCO-08 representations, producing 68.0% exact agreement with institutional SOC–ISCO concordances and consistent results across O*NET releases. The crosswalk allows task-based measures derived from O*NET—including indices of automation exposure, skill content, and recent LLM-based occupational assessments—to be applied directly to internationally comparable labour statistics organised under ISCO-08. This enables cross-country analysis of technological change at the task level without requiring occupation-level aggregation prior to translation between classification systems. One limitation is structural: because O*NET reflects the U.S. occupational structure, ISCO groups without a functional U.S. counterpart cannot be matched directly. In practice, this affects a small number of cases, most notably ISCO 7516 (Tobacco Preparers and Tobacco Products Makers). The crosswalk dataset and replication materials are provided in the Online Appendix and a public data repository.

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Data Availability

The crosswalk dataset and replication materials are provided in the online supplementary material and will be deposited in a public repository upon acceptance.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic), ChatGPT (OpenAI), Claude Code (Anthropic), and Codex (OpenAI) for language editing and writing assistance, and for code generation in the development of the text similarity pipeline. After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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